8. FOURIER TRANSFORM AND FREQUENCY DOMAIN ANALYSIS

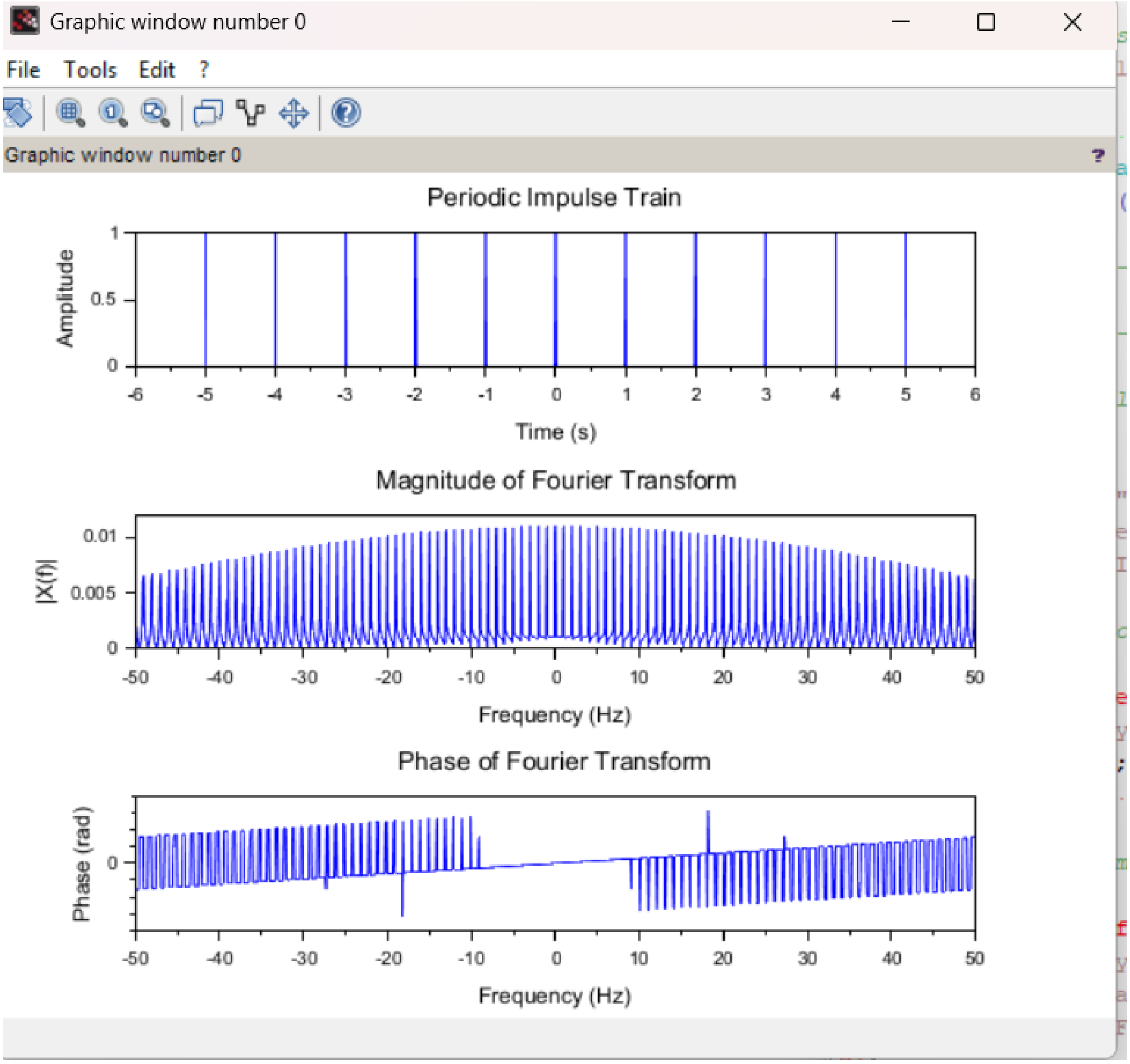
Experiment 8.1

```
clc;
clear;
clf;

//============================================
// Fourier Transform of a Periodic Impulse Train
//============================================

// Time Parameters
t_min = -5;
t_max = 5;
dt = 0.01;
t = t_min:dt:t_max;

//----------------------------------------
// Periodic Impulse Train
//----------------------------------------

T = 1;      // Period
A = 1;      // Amplitude
```

```
x_t = zeros(t);

for k = 1:length(t)
    if abs(modulo(t(k), T)) < dt/2 then
        x_t(k) = A;
    end
end

//----------------------------------------
// Fourier Transform using FFT
//----------------------------------------

N = length(x_t);

X_f = fft(x_t, -1);

X_f_shifted = fftshift(X_f);

// Frequency Axis
f = (-N/2 : N/2-1) / (N*dt);

// Magnitude and Phase
magnitude_X_f = abs(X_f_shifted)/N;
phase_X_f = atan(imag(X_f_shifted), real(X_f_shifted));
```

```
//----------------------------------------
// Plotting
//----------------------------------------

// Periodic Impulse Train
subplot(3,1,1);
plot(t, x_t);
xlabel("Time (s)");
ylabel("Amplitude");
title("Periodic Impulse Train");

// Magnitude Spectrum
subplot(3,1,2);
plot(f, magnitude_X_f);
xlabel("Frequency (Hz)");
ylabel("|X(f)|");
title("Magnitude of Fourier Transform");

// Phase Spectrum
subplot(3,1,3);
plot(f, phase_X_f);
xlabel("Frequency (Hz)");
ylabel("Phase (rad)");
```

title("Phase of Fourier Transform");

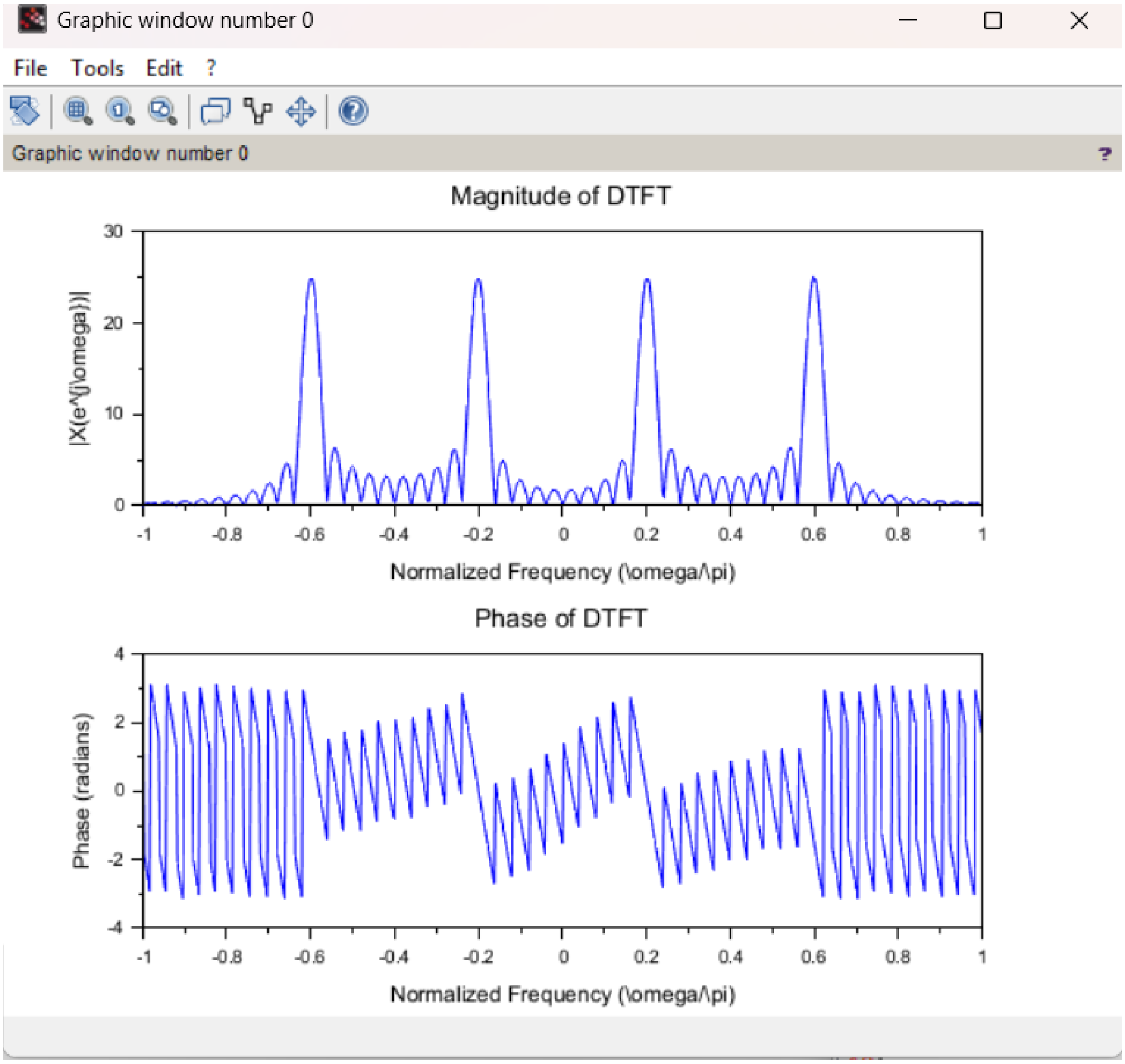


Experiment 8.2

```
clc;
clear;
clf;

//========================================
// DTFT of a Discrete-Time Signal
//========================================

// Parameters
N = 50;
f0 = 0.1;
f1 = 0.3;

// Sample Index
n = 0:N-1;

// Signal
x = cos(2*%pi*f0*n) + sin(2*%pi*f1*n);

// Frequency Range
omega = -%pi:0.01:%pi;
M = length(omega);
```

```
// Compute DTFT
X = zeros(1, M);

for k = 1:M
   X(k) = sum(x .* exp(-%i * omega(k) * n));
end

//----------------------------------------
// Magnitude Plot
//----------------------------------------
subplot(2,1,1);
plot(omega/%pi, abs(X));
xlabel("Normalized Frequency (\omega/\pi)");
ylabel("|X(e^{j\omega})|");
title("Magnitude of DTFT");

//----------------------------------------
// Phase Plot
//----------------------------------------
subplot(2,1,2);
plot(omega/%pi, atan(imag(X), real(X)));
xlabel("Normalized Frequency (\omega/\pi)");
ylabel("Phase (radians)");
```

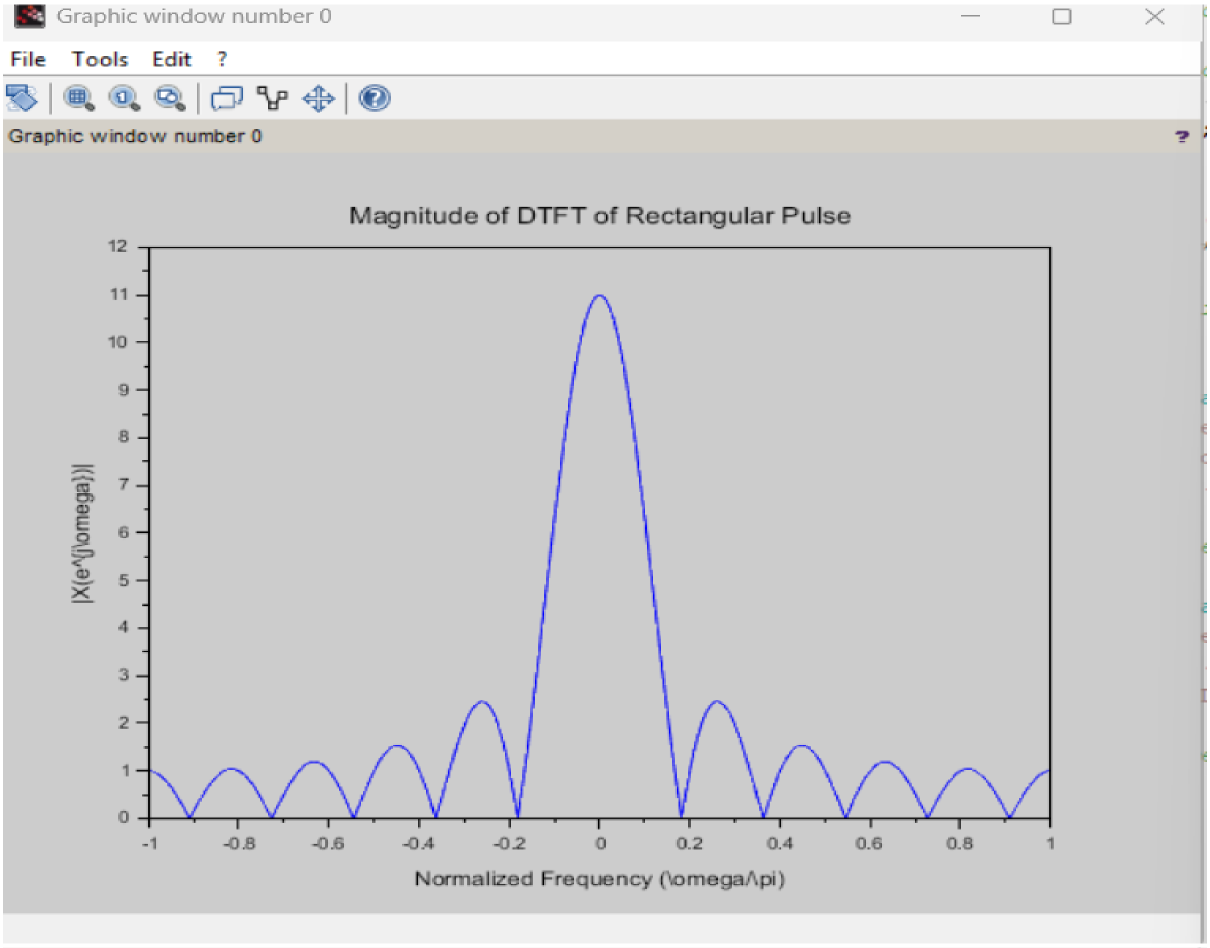
```
title("Phase of DTFT");
```

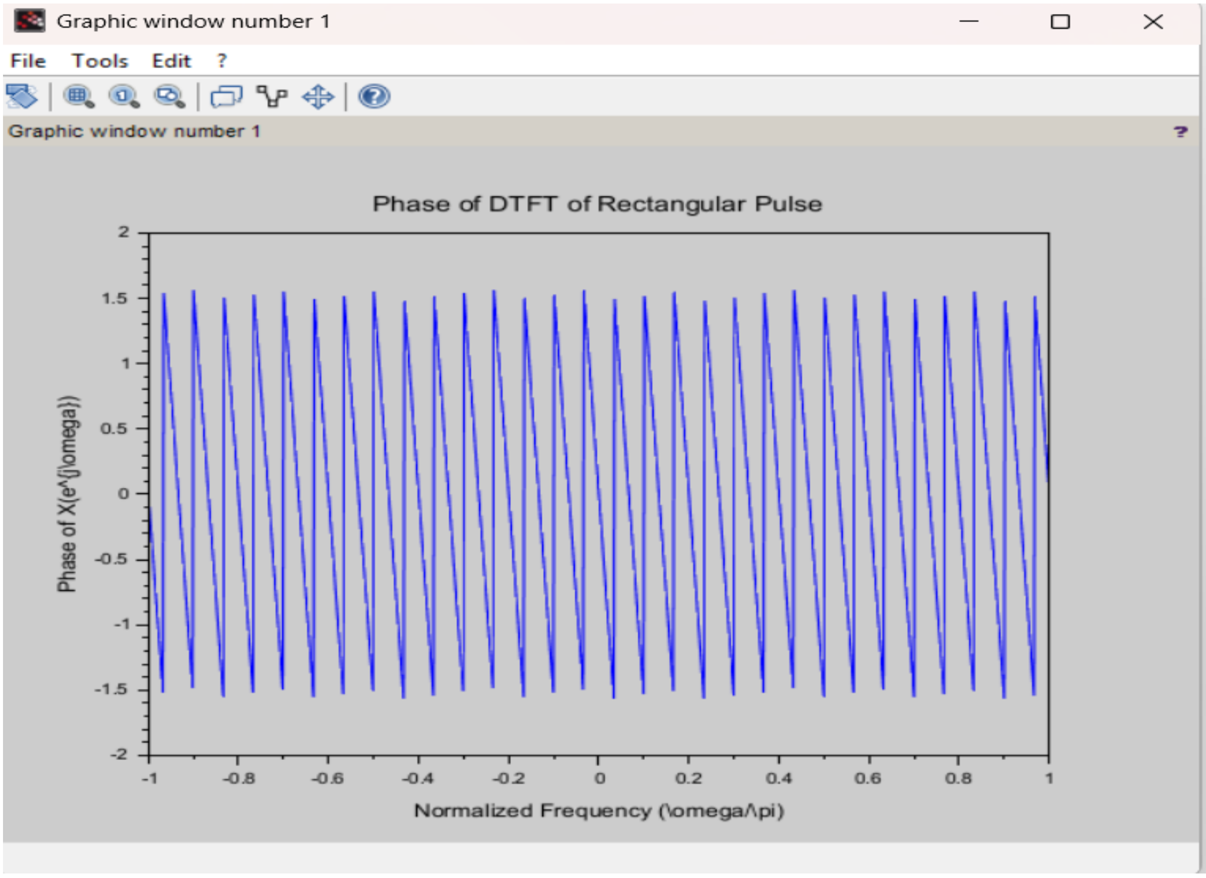


Experiment 8.3

```
// Define parameters
N = 31; // Total number of samples
n = 0:N-1; // Sample index
n1 = 10; // Start of rectangular pulse
n2 = 20; // End of rectangular pulse


// Define the rectangular pulse x[n]
```

```
x = zeros(1, N); // Initialize the signal with zeros
x(n1+1:n2+1) = 1; // Set values between n1 and n2 to 1

// Compute DTFT
omega = linspace(-%pi, %pi, 1000); // Frequency range
X = x * exp(-%i * n' * omega); // DTFT calculation

// Plot the magnitude of DTFT


figure();
plot(omega/%pi, abs(X));
xlabel('Normalized Frequency (\omega/\pi)');
ylabel('|X(e^{j\omega})|');
title('Magnitude of DTFT of Rectangular Pulse');

// Plot the phase of DTFT
figure();
plot(omega/%pi, atan(imag(X)./real(X)));
xlabel('Normalized Frequency (\omega/\pi)');
ylabel('Phase of X(e^{j\omega})');
title('Phase of DTFT of Rectangular Pulse');

// Analyze the results in the Scilab console or report\
```

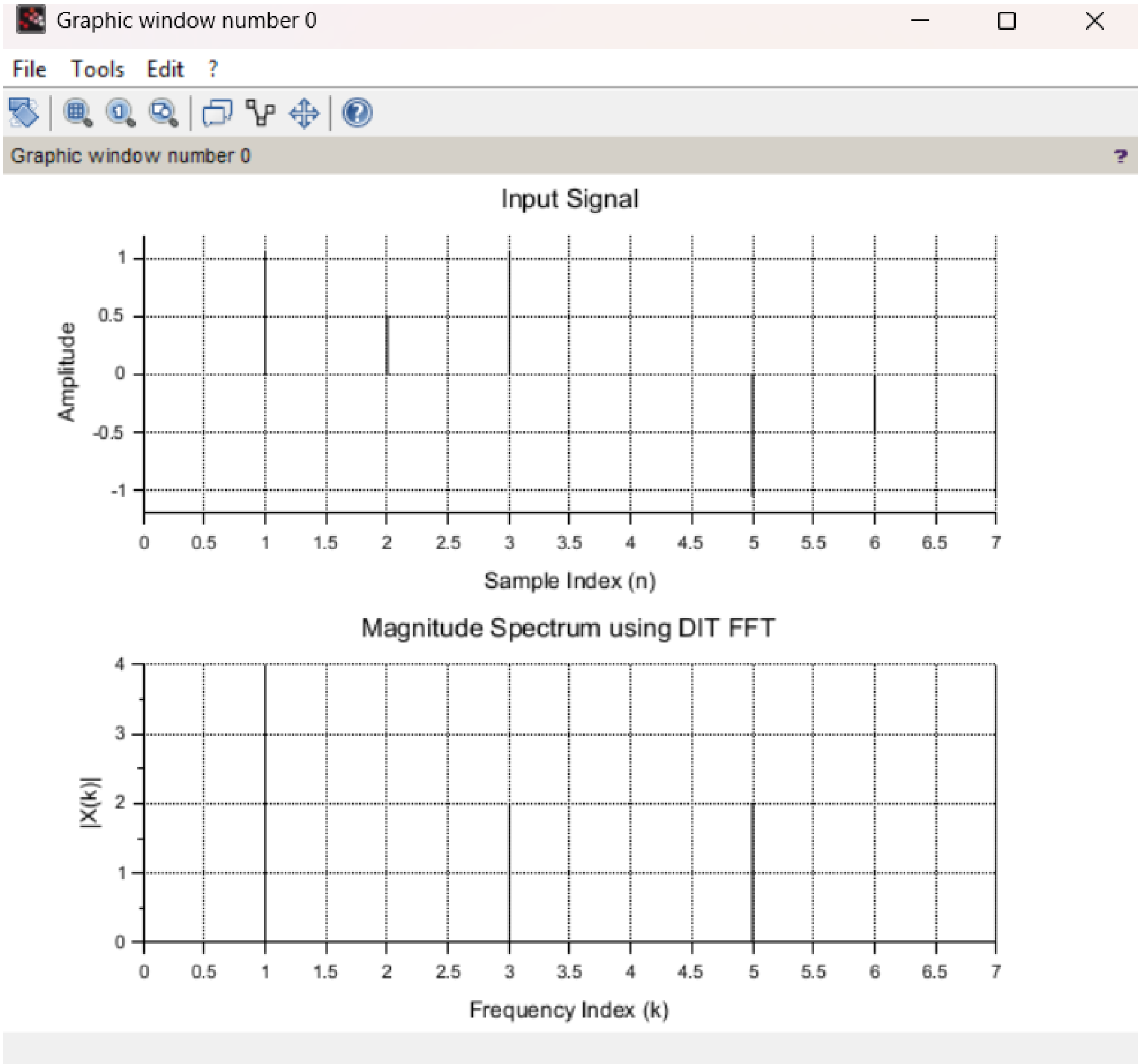

Graphic window number 0
File Tools Edit ?
Graphic window number 0
Magnitude of DTFT of Rectangular Pulse
|X(e^{j\omega})|
Normalized Frequency (\omega/\pi)


Graphic window number 1
File Tools Edit ?
Graphic window number 1
Phase of DTFT of Rectangular Pulse
Phase of X(e^{j\omega})
Normalized Frequency (\omega/\pi)

Experiment 8.4

```
clc;
clear;
clf;

//==========================================
// DIT FFT Implementation
//==========================================

// Input Signal
N = 8;                  // Number of samples
n = 0:N-1;

// Signal
x = sin(2*%pi*n/N) + 0.5*sin(2*%pi*3*n/N);

//-------------------------------------
// FFT Computation
//-------------------------------------
X_fft = fft(x,-1);

// Magnitude Spectrum
magX = abs(X_fft);
```

```
//--------------------------------------
// Plot Input Signal
//--------------------------------------
subplot(2,1,1);
plot2d3(n, x);
xlabel("Sample Index (n)");
ylabel("Amplitude");
title("Input Signal");
xgrid();

//--------------------------------------
// Plot Magnitude Spectrum
//--------------------------------------
subplot(2,1,2);
plot2d3(n, magX);
xlabel("Frequency Index (k)");
ylabel("|X(k)|");
title("Magnitude Spectrum using DIT FFT");
xgrid();
```


Graphic window number 0
File
Tools
Edit
?
Graphic window number 0
Input Signal
Amplitude
1
0.5
0
-0.5
-1
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
5.5
6
6.5
7
Sample Index (n)
Magnitude Spectrum using DIT FFT
|X(k)|
4
3
2
1
0
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
5.5
6
6.5
7
Frequency Index (k)